

BHARATHRAJ R

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PROFESSIONAL SUMMARY

Pre-final year Biomedical Engineering student with strong analytical and problem-solving abilities, backed by hands-on project experience in technology-driven solutions. Dedicated, detail-oriented, and quick to adapt, with proven teamwork and communication skills. Eager to apply knowledge in a dynamic environment, contribute to organizational success, and grow as an innovative professional.

TECHNICAL SKILLS

Programming & Tools: C++, Python, Arduino IDE, Replit

Platforms & Software: Arduino, Raspberry Pi, ESP32

Software & Simulation: Proteus, MATLAB, Scilab, VS Code, Git/GitHub, Anaconda

Data & Analytics: MS Excel, Power BI (Data Visualization, Dashboard Creation)

Key Skills: IoT In Embedded Systems, PCB Design, Microcontroller & microprocessor, Image Processing, Cloud Integration, AI-based Applications, No Code App Development, Problem Solving, Project Management

AI TOOL: ChatGPT, Gemini, GitHub Copilot, Bolt AI, Cursor AI, Replit Ghostwriter, JotForm, OpenCV, Hugging Face

EDUCATION

B. E in BIO MEDICAL ENGINEERING | Karpagam Academy of Higher Education, Coimbatore | **2023 – 2027** | **CGPA-7.01**

PROJECTS

Hand Hygiene Compliance System

- Developing a medical device – an AI & IoT-based Hand Hygiene Monitoring System – to detect handwashing and verify cleanliness using smart sensors. Integrated microcontroller processing with cloud reporting for real-time compliance in hospitals and clinics. Focused on reducing hospital-acquired infections while showcasing skills in IoT, embedded systems, and healthcare innovation

AI-powered Health & Wellness Platform (Ykarb)

- Developed a web & mobile app for menstrual health tracking, mental wellness support, and voice-to-text notes. Integrated Bolt AI for cycle prediction, multilingual chatbot & emotional analysis, and used Supabase for authentication and real-time data storage.

Image Upload and Analysis via API

- Built a web-based AI application with Streamlit for image uploads and automated health-related insights. Integrated GoRaq API with CNN-based deep learning models for image classification and diagnostic feature extraction.

PDF QA System using API

- Implemented an AI-based PDF Question-Answering system using GoRaq API. Integrated LLM-driven responses, sidebar for model selection/tuning, and OCR with Deep Learning-based document classifiers for intelligent document analysis.

Mental Health Assessment App

- Developed a no-code mobile application for self-monitoring emotional well-being. Designed to provide low-cost, stigma-free mental health support enabling accessible and user-friendly mental health care.
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INTERSHIP EXPERIENCE

SriVaariScans&MedhealthDiagnostics, Dharmapuri

June 18, 2024 – July 02, 2024

- Acquired hands-on experience with CT, X-RAY, ULTRASOUND referral lab equipment.
- Assisted in troubleshooting and maintaining diagnostic imaging machines.
- Gained insight into radiology workflows and patient diagnostics.
- Worked closely with professionals to understand healthcare technology Contributed to an academic research project involving biomedical diagnostics.

AI and NLP Development Solaxta Bsol Pvt Ltd

November 29, 2024- December 10, 2024

- Improved a medical chatbot using Retrieval-Augmented Generation (RAG).
- Gained experience in NLP, AI models, and chatbot technology for healthcare.
- Focused on enhancing response accuracy and conversational AI effectiveness.
- Explored ML techniques to advance healthcare automation.

Strydo Technologies Pvt. Ltd, Tirupati

April 28, 2025 – May 31, 2025

- Gained hands-on experience in Embedded Systems, IoT, and Raspberry Pi.
- Worked on microcontroller programming, sensor interfacing, and communication protocols.
- Learned to analyse pin diagrams and datasheets for system design.
- Practiced with tools like VNC Viewer and IP Scanner (19c) for real-time applications.

Basic Web Development InternPE (Online)

Dec 2024 – Jan 2025

- Basic experience gained in HTML, CSS, and JavaScript.
 - simple website design in UI/UX and responsiveness.
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CERTIFICATIONS

- **Electroanalytical Techniques for (Bio) Sensing and (Electro) Catalytic Applications** – CSIR-CECRI, Karaikudi (Oct 2024)
- **Biomedical Equipment Training Program** – Koach & Kauvery Hospital (Oct 2024)
- **AI & ML Applications in Electrical and Electronics Engineering** – NITTTR Chennai (Apr–May 2025)
- **Biomedical Instrumentation & Sensors** – SWAYAM–NPTEL, CSVTU Bhilai (May 2025, Secured 70%)
- **PCB Design** – SSD Trace tech, Coimbatore (Mar 2025, 5-Day Training)
- **Power BI** – Mastering Data Visualization – DIGI Skill Development Centre, Mumbai (Jul 2025, 3-Day National Workshop)